

7. Integration Techniques

7.3. Trigonometric Substitutions

7.3a Learning Objectives

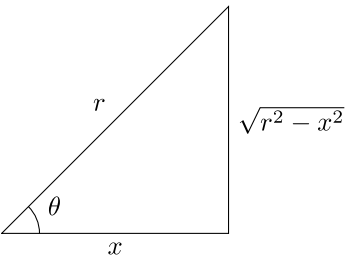
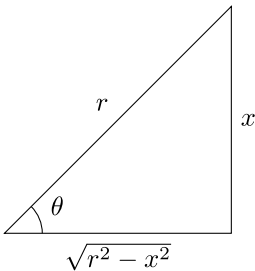
- I can understand the relationship between right triangles and substitutions.
- I can draw triangles to represent substitutions.
- I can understand when and how to use trigonometric substitutions.

Suppose we want to derive the formula for area of a circle of radius r . We know how to set up the integral for this using area between curves:

$$A = \int_{-r}^r \underbrace{\sqrt{r^2 - x^2}}_{\text{top}} - \underbrace{(-\sqrt{r^2 - x^2})}_{\text{bottom}} dx$$

$$= 2 \int_{-r}^r \sqrt{r^2 - x^2} dx$$

Up until now, we don't have a method to evaluate this integral. Looking at a point on the circle we can use trig functions to rewrite the expressions in x and y into r and θ . We have a few options, so let's do both and compare.

Step	Cosine	Sine
substitution	$x = r \cos(\theta)$	$x = r \sin(\theta)$
triangle		
differential	$dx = -r \sin(\theta) d\theta$	$dx = r \cos(\theta) d\theta$
bounds for θ	$x = -r \Rightarrow \theta = \pi$ $x = r \Rightarrow \theta = 0$	$x = -r \Rightarrow \theta = 0$ $x = r \Rightarrow \theta = \frac{\pi}{2}$
integral after trig sub steps below!	$2 \int_{\pi}^0 r \sin(\theta) (-r \sin(\theta)) d\theta$	$2 \int_0^{\frac{\pi}{2}} r \cos(\theta) r \cos(\theta) d\theta$
integral after trig identity	$2r^2(-1) \int_{\pi}^0 \frac{1}{2}(1 - \cos(2\theta)) d\theta$	$2r^2 \int_0^{\frac{\pi}{2}} \frac{1}{2}(1 + \cos(2\theta)) d\theta$
Observations	• bounds are flipped • extra negatives	• the bounds are least to greatest • there are no extra negatives

For $x = r \sin(\theta)$ we get $dx = r \cos(\theta) d\theta$. We have to solve little equations for θ to get our new bounds. Then the integral becomes

$$\begin{aligned}
 A &= 2 \int_{-r}^r \sqrt{r^2 - x^2} dx \\
 &= 2 \int_0^{\frac{\pi}{2}} \sqrt{r^2 - (r \sin(\theta))^2} (r \cos(\theta)) d\theta \\
 &= 2 \int_0^{\frac{\pi}{2}} \sqrt{r^2 - r^2 \sin^2(\theta)} (r \cos(\theta)) d\theta \\
 &= 2 \int_0^{\frac{\pi}{2}} \sqrt{r^2(1 - \sin^2(\theta))} (r \cos(\theta)) d\theta \\
 &= 2 \int_0^{\frac{\pi}{2}} \sqrt{r^2 \cos^2(\theta)} (r \cos(\theta)) d\theta \\
 &= 2 \int_0^{\frac{\pi}{2}} r |\cos(\theta)| (r \cos(\theta)) d\theta \\
 &= 2 \int_0^{\frac{\pi}{2}} r^2 \cos^2(\theta) d\theta \\
 &= 2 \int_0^{\frac{\pi}{2}} r^2 \left[\frac{1}{2} (1 + \cos(2\theta)) \right] d\theta \\
 &= r^2 \int_0^{\frac{\pi}{2}} (1 + \cos(2\theta)) d\theta
 \end{aligned}$$

Ultimately, either substitution **could** work **if you are careful enough**. However, we will just show the substitutions and patterns for the trig subs that don't give extra minus signs. If you are still confused on why we do things in the order we do them, please come chat in office hours about it! (It honestly took me a while to figure this out myself!)

could write $\cos(\theta) = \frac{x}{r}$ or $x = r \cos(\theta)$. This gives us

Find the area of a circle with radius r using techniques of integration.

7.3a Section Summary: